25901637 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BCA (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION APRIL 2026

SECOND SEMESTER

Core Course (CC) - MG2CCRBCA101 - DATA STRUCTURES

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer all questions

Each question carries **2** marks

1. Differentiate between best-case and worst-case time complexity of an algorithm. [A] / [CO1]
2. What is the role of the "top" variable in an array-based stack implementation? [U] / [CO2]
3. Explain the differences between singly linked lists and doubly linked lists. [An] / [CO3]
4. Describe a subgraph. How is it derived from a given graph. [U] / [CO4]
5. What is the difference between an input-restricted and an output-restricted Double-ended-queue? [An] / [CO2]

[2x5 = 10]

Part B

Short Answer Questions

Answer any **5** questions

Each question carries **6** marks

6. Write an algorithm to find the largest element in an array and analyze its time complexity. [A] / [CO1]

7. Define postfix notation and explain how it simplifies the evaluation of arithmetic expressions. [K] / [CO2]

8. Write an algorithm for deleting the first and last node in a circular linked list. Explain it using a suitable example. [A] / [CO3]

9. Briefly explain the different types of rotations in an AVL tree. [U] / [CO4]

10. Write a C program to implement a linear search algorithm. How does the dataset size affect its performance? [A] / [CO1]

11. Describe hashing and the different types of mathematical functions that can be used as hash functions. [U] / [CO3]

12. Compare binary trees and binary search trees with examples. [An] / [CO4]

[6x5 = 30]

Part C

Essay Questions

Answer any **2** questions

Each question carries **15** marks

13. Explain how a two-dimensional array is represented in memory. Write a C program to multiply two matrices. [A] / [CO1]

14. Compare simple queues and circular queues with examples and their applications. [An] / [CO2]

15. Explain the different types of recursion? Write a C program to generate fibonacci series upto a limit using recursion. [A] / [CO3]

[15x2 = 30]